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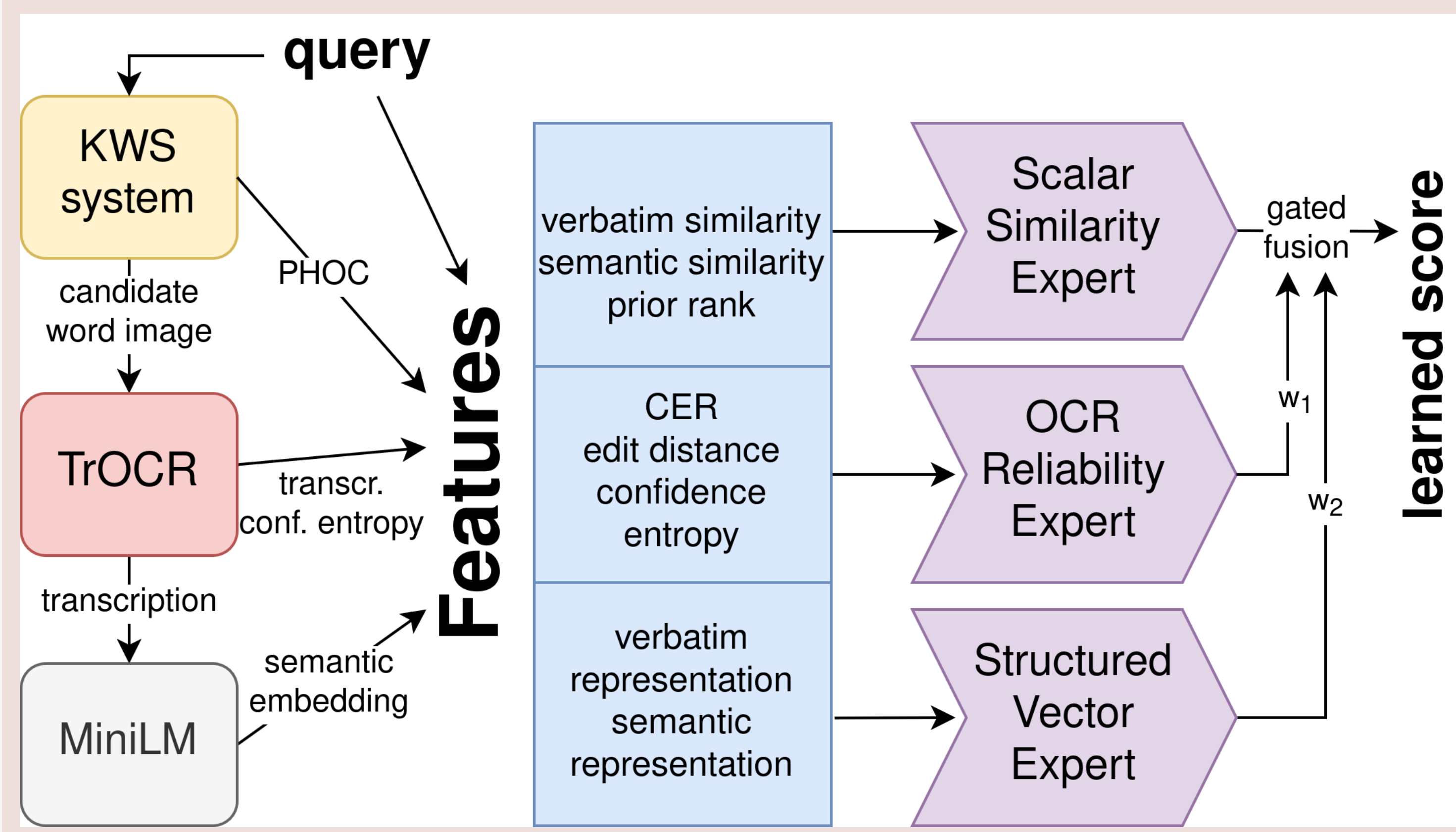
- ▶ Segmentation-free Keyword Spotting¹ (KWS) is an information retrieval paradigm where query-relevant candidates are retrieved **without** explicit word-level segmentation or recognition
- ▶ Enables robust search in historical handwritten documents and other unstructured textual evidence
- ▶ Particularly relevant in forensic document analysis: **degraded**, **unconstrained**, or **partially legible** content
- ▶ Existing methods typically rely on **visual similarity** ignoring semantics or employing trivial **linear** score fusion²

- Move **beyond** visual-only matching by also incorporating semantic and textual information
- Signals encoded as tokens in a shared latent space
- Use query-conditioned cross-attention for **adaptive evidence aggregation** and **multi-modal reasoning**
- Plug-and-play **post-processing module** compatible with existing KWS systems

- ▶ Reformulate KWS re-ranking as **representation-level multi-token fusion** instead of score-level fusion
- ▶ Introduce a **query-conditioned multi-expert attention model** with reliability-aware fusion
- ▶ Achieve **91.2 mAP@25** on IAM, outperforming linear fusion (88.25%) and approaching oracle (93.77%)

- ▶ Initial KWS³ retrieves candidate regions
- ▶ OCR decoder⁴ produces textual transcriptions
- ▶ Semantic embeddings⁵ encode textual meaning
- ▶ Re-ranking model combines visual and textual signals
- ▶ Final ranking refined using learned scores

- ▶ Query-conditioned multi-expert architecture
- ▶ ▷ **Scalar expert:** visual/semantic similarity, prior-rank position
- ▶ ▷ **OCR expert:** OCR confidence, CER, edit distance, entropy
- ▶ ▷ **Vector expert:** PHOC descriptors, semantic embeddings
- ▶ Cross-attention over tokenized representations
- ▶ Scalar expert = **anchor**; complementary experts refine the representation via gated fusion



- ▶ IAM Handwriting Database: standard KWS benchmark, challenging due to high variability (1,539 pages, 657 writers, 115,320 word-level annotations)
- ▶ **Learning-To-Rank**: Pairwise ranking loss with hard negative mining (reducing false positive rankings)
- ▶ Evaluation metric: mAP@25 at 0.25 IoU

Method	mAP@25
Baseline ³	86.40
Linear fusion of scores ²	88.25
Proposed Method	91.20
Oracle (ideal re-reranking)	93.77

► **+4.8 mAP@25** over baseline, **+2.95** over linear fusion, significantly narrowing gap to oracle



- ▶ Token-level fusion outperforms score-level fusion
- ▶ Query conditioning enables adaptive weighting of heterogeneous signals
- ▶ OCR uncertainty provides critical corrective signals
- ▶ Hard negative mining improves ranking robustness
- ▶ Strong improvements on IAM benchmark

- ▶ Re-ranking reformulated as multi-token attention over heterogeneous signals
- ▶ Adaptive multi-expert fusion improves robustness and interpretability
- ▶ Strong gains on IAM benchmark
- ▶ Provides interpretable modality-level evidence attribution

1. Giotis et al., *A Survey of Document Image Word Spotting Techniques*, Pattern Recognition, 2017
2. Papazis et al., *Enhancing Keyword Spotting via NLP-Based Re-Ranking: Leveraging Semantic Relevance Feedback in the Handwritten Domain*, Electronics 2025
3. Retsinas et al., *Keyword Spotting Simplified: A Segmentation-Free Approach Using Character Counting and CTC Re-Scoring*, ICDAR 2023
4. Li et al., *TROCR Transformer-Based Optical Character Recognition with Pre-Trained Models*, AAAI 2023
5. Reimers et al., *Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks*, EMNLP-IJCNLP 2019.

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